

Skill-7

Program:

```
tejaswini@teju: ~/OS_Lab/skil × + ▾
GNU nano 8.7.1 skill_7.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
int main() {
    char *path = getenv("PATH");
    if (path == NULL) {
        printf("PATH variable not found.\n");
        return 1;
    }

    printf("PATH Variable:\n%s\n\n", path);

    pid_t pid = fork();

    if (pid < 0) {
        perror("fork failed");
        return 1;
    }

    if (pid == 0) {
        // Child process
        printf("Child Process Started\n");
        printf("Child PID: %d\n", getpid());

        // Execute ls
        execlp("ls", "ls", NULL);

        // Runs only if exec fails
        perror("exec failed");
        exit(1);
    } else {
        // Parent process
        int status;

        printf("Parent waiting for child...\n");
        waitpid(pid, &status, 0);

        if (WIFEXITED(status)) {
            printf("Child exited with status: %d\n", WEXITSTATUS(status));
        }
    }

    return 0;
}
```

Output:

```
tejaswini@teju:~/OS_Lab/skill$ nano skill_7.c
tejaswini@teju:~/OS_Lab/skill$ gcc skill_7.c -o skill_7
tejaswini@teju:~/OS_Lab/skill$ ./skill_7
PATH Variable:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wnsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux_2
.7.11.0_x64__8wekyb3d8bbwe:/mnt/c/Program Files (x86)/VMware/VMware Workstation/bin:/mnt/c/Program Files/Java/jdk-25/bin:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Prog
ram Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/WINDOWS/system32:/mnt/c/WINDOWS:/mnt/c/WINDOWS/System32/wbem:/mnt/c/W
INDOWS/System32/WindowsPowerShell/vi.8/;/mnt/c/WINDOWS/System32/OpenSSH;/mnt/c/Program Files (x86)/NVIDIA Corporation/PhysX/Common:/mnt/c/Program Files/NVIDIA Corporation/NVIDIA NvDLISR:/
mnt/c/Users/Tejaswini/Downloads/gcc-15.1.0/bin:/mnt/c/Program Files/nodejs/;/mnt/c/Program Files/Git/cmd:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin:/mnt/c/Users/Tejaswini/AppData/Loca
l/Microsoft/WindowsApps:/mnt/c/Users/Tejaswini/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/Tejaswini/AppData/Roaming/npm:/mnt/c/Users/Tejaswini/AppData/Local/Python/bin:/mnt/
c/Program Files/JetBrains/PyCharm 2026.1/bin:/mnt/c/apache-maven-3.9.16-bin/apache-maven-3.9.16/bin:/mnt/c/Program Files/MongoDB/Server/8.3/bin:/mnt/c/Users/Tejaswini/AppData/Local/Program
s/mongodb/snap/bin

Parent waiting for child...
Child Process Started
Child PID: 684
child_process child_process.c escape_sequence escape_sequence.c progl progl.c progl.c.save prog2 prog2.c skill_5 skill_5.c skill_6 skill_6.c skill_7 skill_7.c
Child exited with status: 0
tejaswini@teju:~/OS_Lab/skill$ |
```